

# 543 Readmissions Project Exploratory Data Analysis Report

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# 1 Dataset Overview

The dataset comes from two CSV files, `healthcare_readmissions_dataset_train` and `healthcare_readmissions_dataset_test`, each row consisting of one patient encounter. The training file contains 8,038 rows and 19 columns (including the target variable).

| Variable                          | Description                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------|
| PatientID                         | Unique patient identifier                                                               |
| Age                               | Patient age                                                                             |
| Gender                            | String variable with values "Male" or "Female"                                          |
| Ethnicity                         | String variable with values "Caucasian", "Hispanic", "African American" and "Other"     |
| Hospital ID                       | Hospital identifier with values "Hosp1", "Hosp2", and "Hosp3"                           |
| Height                            | Patient height in meters                                                                |
| Smoker                            | Boolean indicating if patient is a current smoker                                       |
| BMI                               | Patient Body Mass Index                                                                 |
| Weight                            | Patient weight in kg                                                                    |
| Adjusted Weight                   | Health system-specific adjustments to patient weight (in kg)                            |
| Has Diabetes                      | Boolean indicating if patient has diabetes (0 or 1)                                     |
| Has Hypertension                  | Boolean indicating if patient has hypertension (0 or 1)                                 |
| Has Chronic Kidney Disease        | Boolean indicating if patient has chronic kidney disease (0 or 1)                       |
| Living Situation                  | Living arrangement: Lives Alone, Lives with Family, Assisted Living                     |
| Exercise Frequency                | String variable with values "None", "Occasional", or "Regular"                          |
| Diet Type                         | String variable with values "Balanced", "High-fat", "Vegetarian", "Other"               |
| Number of Prior Visits            | Number of previous hospitalizations of the patient                                      |
| Medications Prescribed            | Number of different prescription medications patient is currently taking                |
| Length of Stay                    | Length of the hospital stay in days                                                     |
| Type of Treatment                 | String variable with values "None", "Minor Surgery", "Major Surgery", "Other Treatment" |
| Readmission within 30 days        | Binary variable (1 = patient was readmitted, 0 = patient was not readmitted)            |
| Post-Discharge Satisfaction Score | Survey done 30 days after discharge. 1.0 – 5.0 scale                                    |

Figure 1: Data Dictionary

It is worth noting that `Has Chronic Kidney Disease`, `Post-Discharge Satisfaction Score`, and `Living Situation` are not actually present in the files, despite being in the dictionary.

## 2 Data Quality Summary

Only two features in the dataset have missing values: `Number of Prior Visits` (314 rows or 3.9%) and `Medications Prescribed` (657 rows or 8.2%). The remaining features are complete, with no duplicates in the training set either. Although boolean columns are not consistent in datatype (`Smoker` is a boolean column while `Has Diabetes` and `Has Hypertension` are integers with values of 0 and 1), this can be handled quite easily during preprocessing.

`Exercise Frequency` and `Type of Treatment` also feature 'None' as a possible value, which the `pandas` function `pd.read_csv()` will interpret as a `NaN` by default. Again, this is not a major issue, just something to be aware of.

A glance at the distribution of variables shows that `Age` has problematic outliers, in the form of patients over the age of 100.

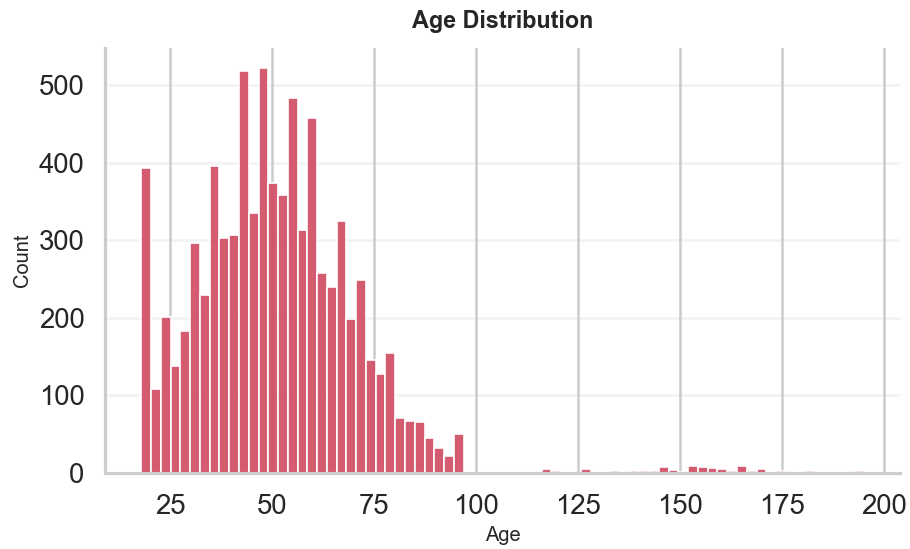


Figure 2: Raw Age Distribution

Using z-scores to identify outliers (as well as common sense), 80 rows were determined to be outliers and subsequently removed from the dataset. These rows make up 0.99% of the training set; although I would prefer to keep as much data as possible, these rows are clearly unrepresentative of real patients (for whatever reason), and could potentially skew the results of any analysis or modeling.

### 3 Statistical Summary

All stats are derived from the training set with the age outliers removed as outlined in the previous section.

#### 3.1 Numeric Statistics

Table 1: Measure Summary Statistics

|                            | count       | mean      | std       | min       | 25%       | 50%       | 75%       | max        |
|----------------------------|-------------|-----------|-----------|-----------|-----------|-----------|-----------|------------|
| age                        | 7958.000000 | 50.110000 | 17.300000 | 18.000000 | 38.000000 | 50.000000 | 62.000000 | 95.000000  |
| height_m                   | 7958.000000 | 1.700000  | 0.100000  | 1.300000  | 1.600000  | 1.700000  | 1.800000  | 2.000000   |
| bmi                        | 7958.000000 | 26.260000 | 4.750000  | 8.300000  | 23.100000 | 26.200000 | 29.300000 | 44.000000  |
| weight_kg                  | 7958.000000 | 77.150000 | 18.950000 | 23.300000 | 64.800000 | 75.400000 | 87.300000 | 236.300000 |
| adjusted_weight_kg         | 7958.000000 | 76.270000 | 16.660000 | 23.130000 | 64.730000 | 75.130000 | 86.860000 | 159.050000 |
| has_diabetes               | 7958.000000 | 0.130000  | 0.340000  | 0.000000  | 0.000000  | 0.000000  | 0.000000  | 1.000000   |
| has_hypertension           | 7958.000000 | 0.170000  | 0.380000  | 0.000000  | 0.000000  | 0.000000  | 0.000000  | 1.000000   |
| number_of_prior_visits     | 7646.000000 | 3.040000  | 1.740000  | 0.000000  | 2.000000  | 3.000000  | 4.000000  | 11.000000  |
| medications_prescribed     | 7308.000000 | 3.510000  | 1.960000  | 0.000000  | 2.000000  | 3.000000  | 5.000000  | 12.000000  |
| length_of_stay             | 7958.000000 | 2.540000  | 2.970000  | 0.000000  | 0.000000  | 2.000000  | 4.000000  | 23.000000  |
| readmission_within_30_days | 7958.000000 | 0.170000  | 0.380000  | 0.000000  | 0.000000  | 0.000000  | 0.000000  | 1.000000   |

#### 3.2 Categorical Statistics



Figure 3: Category Distributions

Individual distributions can be found in the **Univariate Analysis** section below.

## 4 Univariate Analysis

### 4.1 Numeric Features

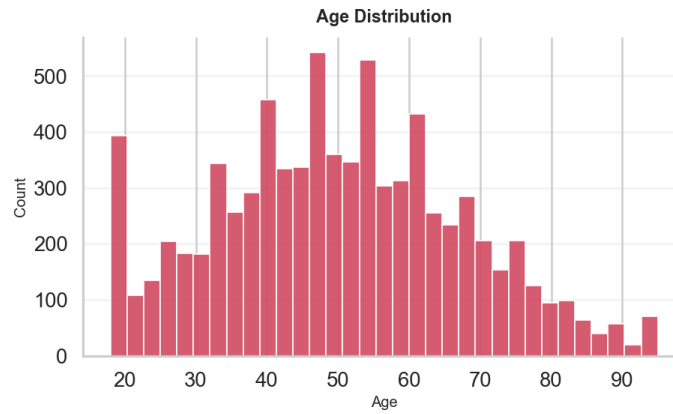


Figure 4: Age Histogram

The distribution with removed outliers makes much more sense compared to the raw values.

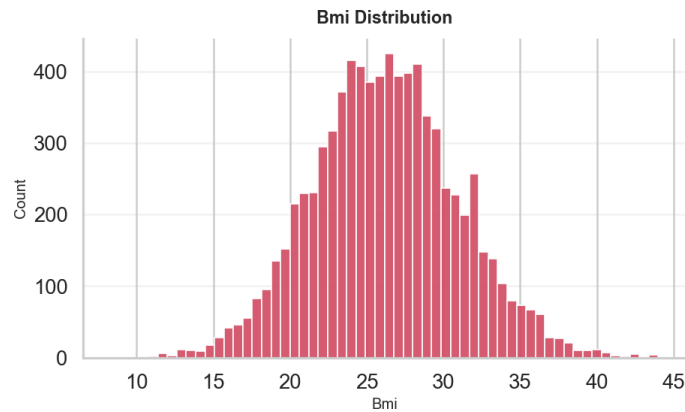


Figure 5: BMI Histogram

For the most part normal; some values on the ends are in the extreme but not necessarily impossible (e.g. a BMI of 44 is very high but not unheard of).

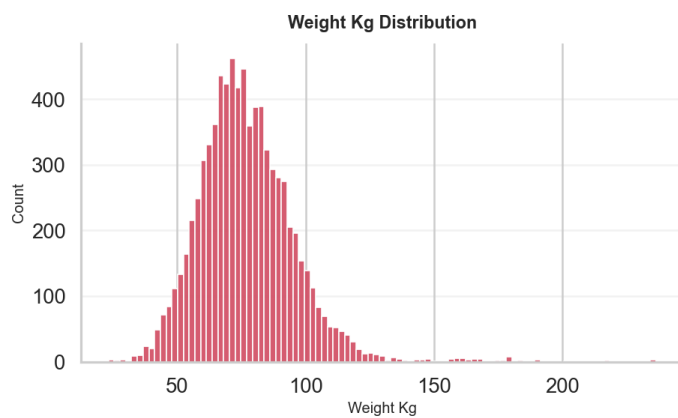


Figure 6: Weight Histogram

Similar to BMI, some extreme values but nothing that stands out as impossible. Also makes sense that its similar to BMI as they are highly correlated.

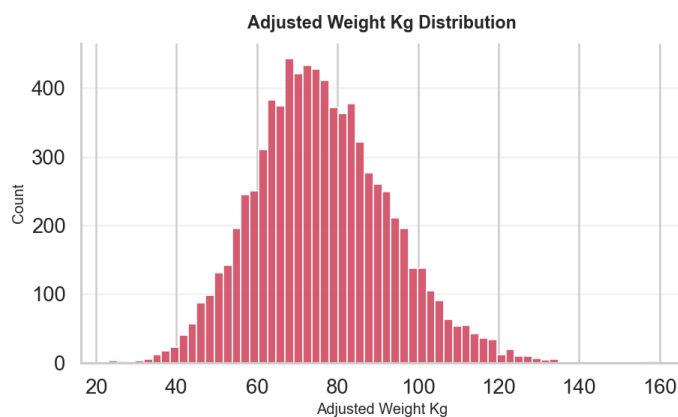


Figure 7: Adjusted Weight Histogram

Less outliers than standard weight, however the Data Dictionary does not specify how (or why) its adjusted, so it's hard to make any judgments or comparisons to standard weight.

## 4.2 Categorical Features

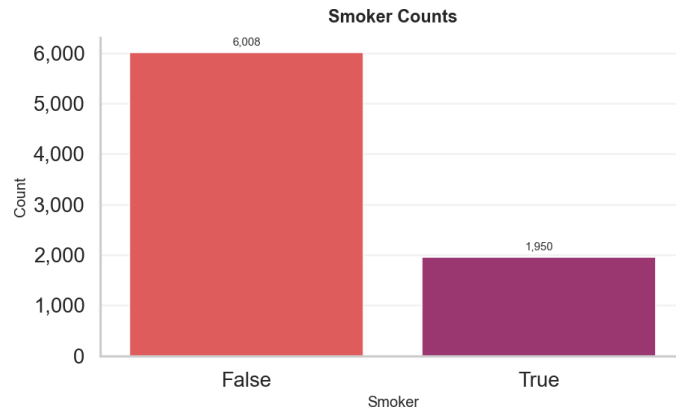


Figure 8: Smoker Bar Chart

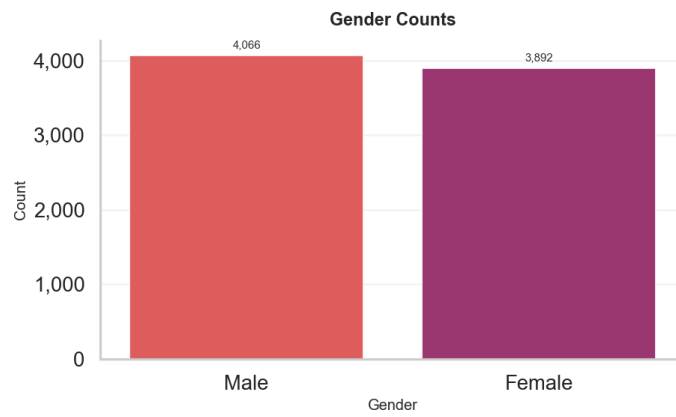


Figure 9: Gender Bar Chart

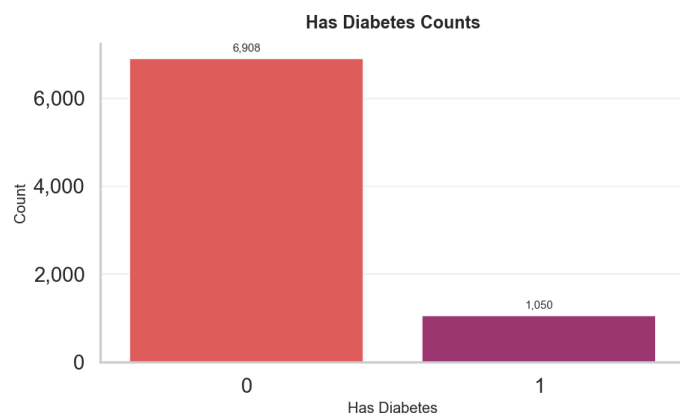


Figure 10: Diabetes Bar Chart

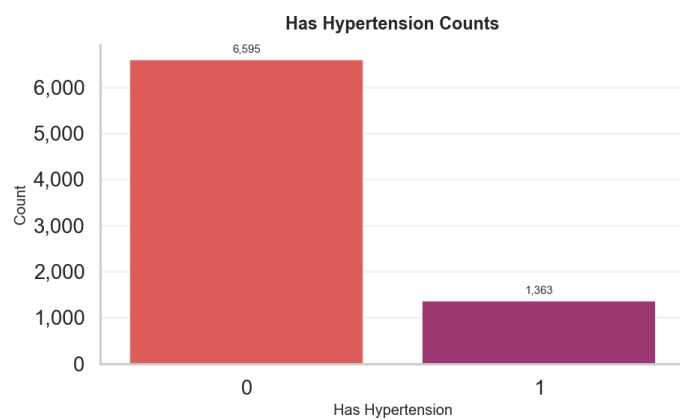


Figure 11: Hypertension Bar Chart



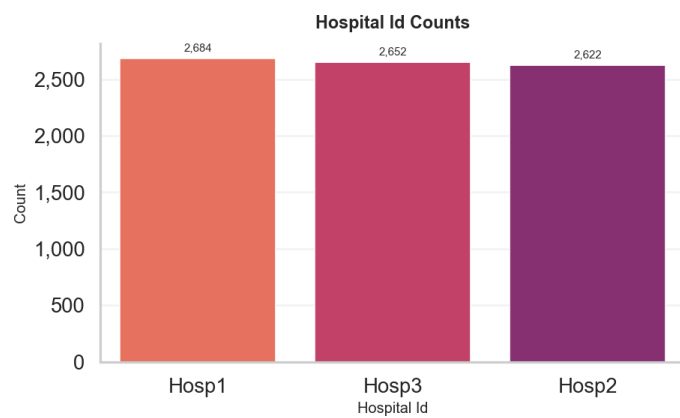


Figure 12: Hospital ID Bar Chart

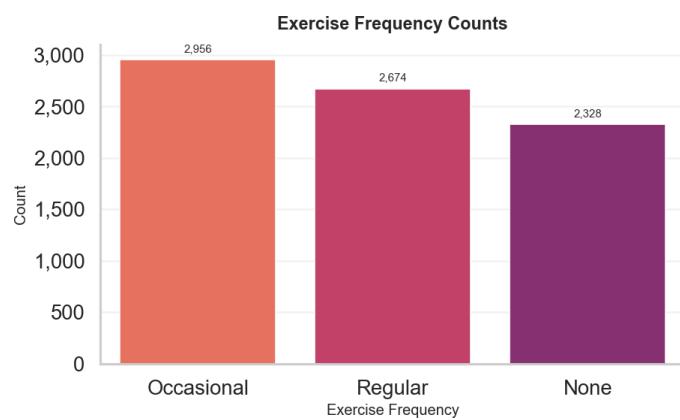


Figure 13: Exercise Frequency Bar Chart

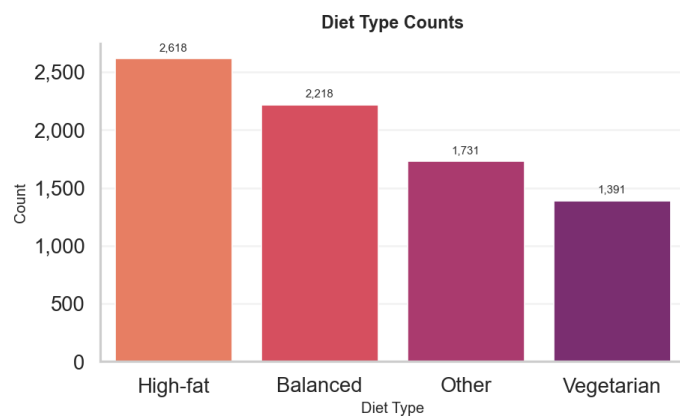


Figure 14: Diet Bar Chart

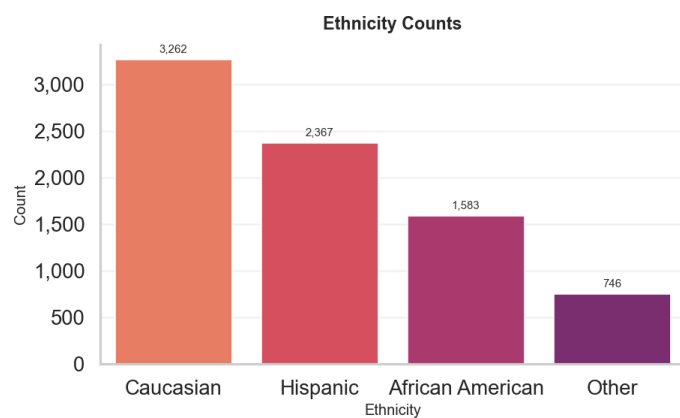


Figure 15: Ethnicity Bar Chart

Figures 8 through 15 are all quite unremarkable and to be expected for the dataset.

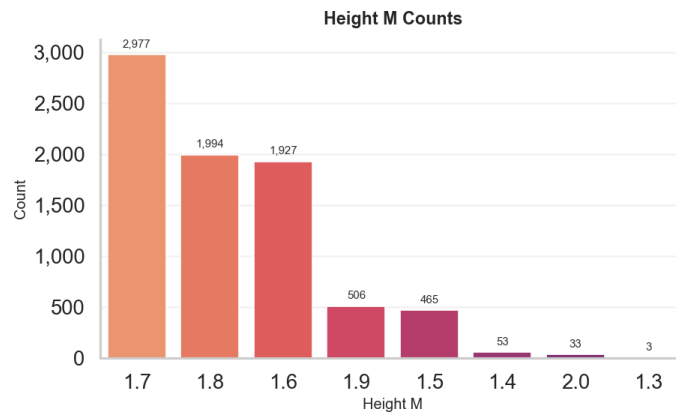


Figure 16: Height Bar Chart

Also expected, although interesting that the height is stored in what is essentially categorical form, rather than more precise measurements.

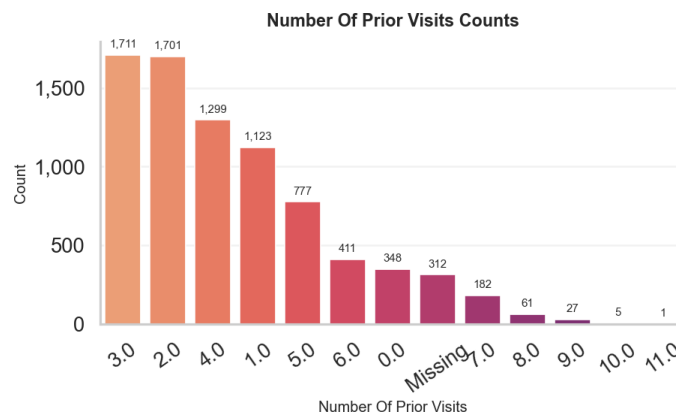


Figure 17: Prior Visits Bar Chart

Has a very long tail, and also missing values are quite prevalent. Would potentially make sense to group/encode values somehow, but that will be looked into during modeling and feature engineering.

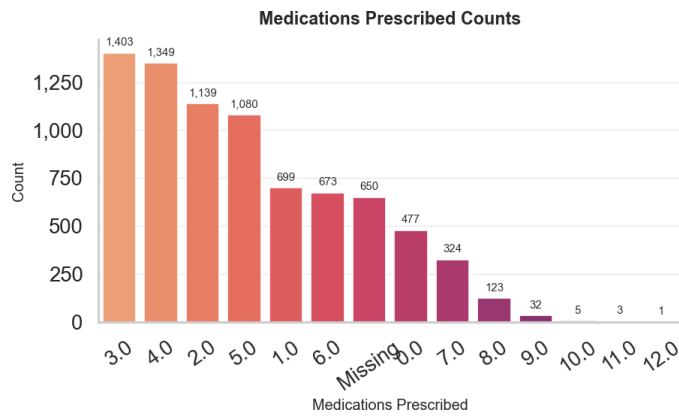


Figure 18: Medications Prescribed Bar Chart

A somewhat similar situation to prior visits, although an overall smaller range of values. Maybe worth grouping into ranges, or just full binary of 0 vs. 1 or more prescriptions.

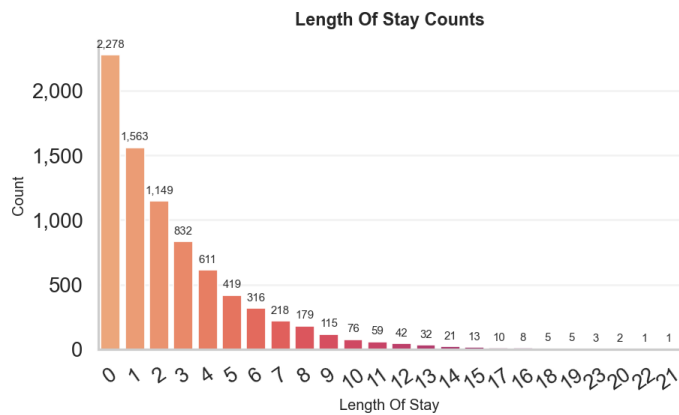


Figure 19: Length of Stay Bar Chart

Long tail, no missing values. Should be encoded somehow, should look to common techniques other projects/models have used in the past as it seems like a common and important feature in healthcare datasets.

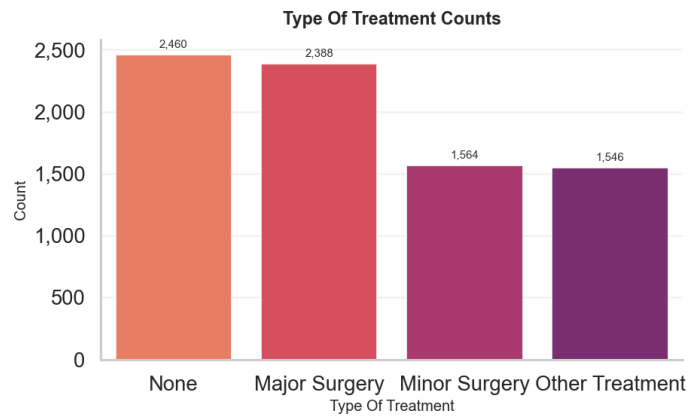


Figure 20: Treatment Type Bar Chart

Somewhat vague categories, more experimentation on encoding strategies should be revealing as per best utilizing the feature.

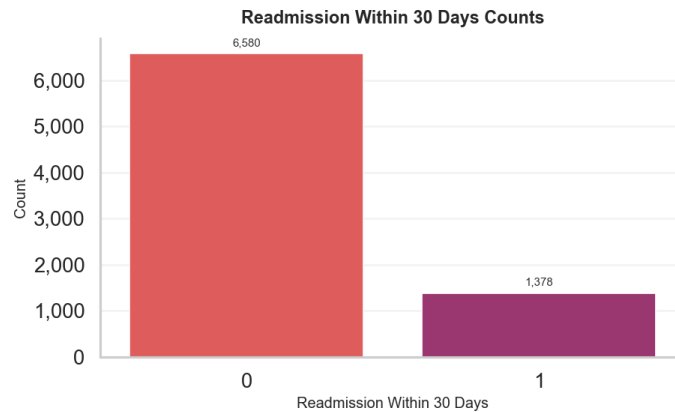


Figure 21: Readmission (Target) Bar Chart

Output variable. Very skewed, should consider that in model selection and/or resampling techniques.

## 5 Bivariate Analysis

### 5.1 Feature vs. Target

### 5.2 Feature Correlations